

ARM Exam Atlas - Master Reference

Searchable exam, solution-pattern, adaptation, and verification reference

How to use the atlas

Start with Quick Pattern Lookup. Search exact normalized tags, then move from pattern to related exams and the complete project.

Coverage

48 question requirements, 23 exam variants, 23 solved projects, 28 stable solution patterns.

Verification meaning

COMPILE_ONLY means no execution claim. SIMULATOR_EXECUTED_PASS requires actual linked ARM execution. HARDWARE_BUILD_PASS is a separate LPC1768 build result. PHYSICAL_BOARD_NOT_TESTED is expected while no board is available.

Pattern frequency

abi:nonleaf: 32 question(s) - PAT-AAPCS-NONLEAF-001
risk:stack-alignment: 28 question(s) - PAT-AAPCS-STACK-SAFETY-001
mem:matrix-row-major: 21 question(s) - PAT-MEM-MATRIX-ROW-MAJOR-001
flow:nested-loop: 20 question(s) - PAT-FLOW-NESTED-LOOP-001
risk:irq-shared-state: 20 question(s) - PAT-STATE-IRQ-HANDOFF-001
board:gpio: 19 question(s) - PAT-GPIO-EVENT-001
alg:recurrence: 18 question(s) - PAT-ALG-RECURRENCE-001
board:timer: 17 question(s) - PAT-TIMER-OWNERSHIP-001
timing:periodic: 15 question(s) - PAT-TIMER-PERIODIC-001
state:event-loop: 14 question(s) - PAT-STATE-EVENT-LOOP-001
cpu:flags: 13 question(s) - PAT-CPU-FLAGS-001
mem:word-array: 13 question(s) - PAT-MEM-WORD-ARRAY-001
mem:byte-array: 11 question(s) - PAT-MEM-BYTE-ARRAY-001
state:debounce: 10 question(s) - PAT-STATE-DEBOUNCE-001
board:adc: 10 question(s) - PAT-ADC-SAMPLE-001
board:dac: 10 question(s) - PAT-DAC-STREAM-001
timing:free-running: 9 question(s) - PAT-TIMER-FREE-RUNNING-001
alg:graph-search: 7 question(s) - PAT-ALG-GRAPH-SEARCH-001
alg:frequency-count: 6 question(s) - PAT-ALG-FREQUENCY-COUNT-001
flow:early-break: 5 question(s) - PAT-FLOW-EARLY-BREAK-001
abi:stacked-args: 5 question(s) - PAT-AAPCS-STACKED-ARGS-001
abi:four-register-args: 4 question(s) - PAT-AAPCS-FOUR-ARGS-001
alg:fixed-point: 4 question(s) - PAT-ALG-FIXED-POINT-001
board:joystick: 4 question(s) - PAT-GPIO-JOYSTICK-001
alg:sorting: 3 question(s) - PAT-ALG-SORTING-001
cpu:svc: 2 question(s) - PAT-CPU-SVC-001
risk:vector-ownership: 2 question(s) - PAT-TIMER-VECTOR-OWNERSHIP-001
cpu:exception-frame: 1 question(s) - PAT-CPU-EXCEPTION-FRAME-001

Exam chronology

E2023-02-07 (ARM): abi:nonleaf, alg:sorting, board:gpio, board:timer, cpu:svc, flow:early-break, flow:nested-loop, mem:byte-array, risk:irq-shared-state, risk:stack-alignment, risk:vector-ownership, state:debounce, timing:free-running

E2023-02-24 (ARM): abi:nonleaf, alg:recurrence, alg:sorting, cpu:exception-frame, cpu:svc, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:stack-alignment

E2023-05-17 (ARM): abi:nonleaf, board:adc, cpu:flags, flow:nested-loop, mem:matrix-row-major, risk:stack-alignment

E2023-07-04 (ARM): abi:nonleaf, alg:recurrence, board:gpio, board:timer, flow:nested-loop, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:event-loop, timing:periodic

E2023-09-18 (ARM): abi:nonleaf, alg:recurrence, board:gpio, cpu:flags, flow:nested-loop, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop

E2024-02-12 (ARM): abi:nonleaf, alg:graph-search, board:gpio, board:timer, cpu:flags, flow:early-break, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop, timing:free-running

E2024-02-28 (ARM): abi:nonleaf, alg:graph-search, alg:recurrence, board:gpio, board:timer, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, state:event-loop, timing:periodic

E2024-07-09 (ARM): abi:four-register-args, abi:nonleaf, alg:graph-search, alg:recurrence, board:timer, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:stack-alignment, timing:periodic

E2024-09-16 (ARM): abi:four-register-args, abi:nonleaf, abi:stacked-args, alg:graph-search, board:gpio, cpu:flags, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop

E2025-01-29-A1 (ARM1): abi:nonleaf, alg:fixed-point, board:gpio, board:timer, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, timing:free-running, timing:periodic

E2025-01-29-A2 (ARM2): board:gpio, board:timer, cpu:flags, flow:early-break, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, timing:free-running, timing:periodic

E2025-01-29-A3 (ARM3): abi:nonleaf, board:dac, board:gpio, board:timer, flow:nested-loop, mem:byte-array, mem:matrix-row-major, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop, timing:free-running, timing:periodic

E2025-02-12-A1 (ARM1): abi:nonleaf, alg:fixed-point, alg:recurrence, board:adc, board:dac, board:gpio, board:timer, cpu:flags, mem:matrix-row-major, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:debounce, timing:periodic

E2025-02-12-A2 (ARM2): abi:nonleaf, alg:fixed-point, alg:recurrence, board:adc, board:dac, board:gpio, board:timer, mem:word-array, risk:irq-shared-state, risk:stack-alignment, timing:periodic

E2025-07-01-A1 (ARM1): abi:nonleaf, abi:stacked-args, alg:recurrence, board:gpio, board:joystick, board:timer, cpu:flags, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:event-loop, timing:periodic

E2025-07-01-A2 (ARM2): abi:nonleaf, abi:stacked-args, alg:recurrence, board:gpio, board:joystick, board:timer, flow:nested-loop, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:event-loop

E2026-02-03-A1 (ARM1): abi:nonleaf, alg:recurrence, board:adc, board:dac, board:gpio, cpu:flags, mem:byte-array, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop

E2026-02-03-A2 (ARM2): abi:nonleaf, board:adc, board:dac, board:gpio, flow:early-break, mem:byte-array, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:debounce

E2026-02-03-A3 (ARM3): abi:nonleaf, alg:recurrence, board:adc, board:dac, board:gpio, board:timer, cpu:flags, flow:nested-loop, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:event-loop, timing:periodic

E2026-02-18-A1 (ARM1): abi:nonleaf, alg:frequency-count, alg:recurrence, board:adc, board:dac, board:gpio, board:timer, mem:word-array, risk:irq-shared-state, risk:stack-alignment, risk:vector-ownership, state:event-loop, timing:free-running, timing:periodic

E2026-02-18-A2 (ARM2): abi:nonleaf, alg:frequency-count, alg:recurrence, board:adc, board:dac, board:timer, cpu:flags, mem:matrix-row-major, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:event-loop, timing:free-running, timing:periodic

E2026-06-25-B1 (ARM1): abi:four-register-args, abi:nonleaf, alg:frequency-count, board:adc, board:gpio, board:joystick, board:timer, mem:matrix-row-major, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop, timing:free-running, timing:periodic

E2026-06-25-B2 (ARM2): abi:four-register-args, abi:nonleaf, alg:frequency-count, board:gpio, board:joystick, board:timer, flow:early-break, flow:nested-loop, mem:word-array, risk:irq-shared-state, risk:stack-alignment, state:debounce, state:event-loop, timing:free-running, timing:periodic

Solution patterns

PAT-ALG-SORTING-001 - Signed and unsigned array sorting. Exams: E2023-02-07, E2023-02-24

PAT-ALG-RECURRENCE-001 - Bounded recurrence into an array. Exams: E2023-02-24, E2023-07-04, E2023-09-18, E2024-02-28, E2024-07-09, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2

PAT-ALG-GRAPH-SEARCH-001 - Matrix and graph traversal. Exams: E2024-02-12, E2024-02-28, E2024-07-09, E2024-09-16

PAT-ALG-FREQUENCY-COUNT-001 - Frequency-count comparison. Exams: E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-ALG-FIXED-POINT-001 - Fixed-point multiply and recurrence. Exams: E2025-01-29-A1, E2025-02-12-A1, E2025-02-12-A2

PAT-FLOW-NESTED-LOOP-001 - Nested loops with preserved outer state. Exams: E2023-02-07, E2023-02-24, E2023-05-17, E2023-07-04, E2023-09-18, E2024-02-12, E2024-02-28, E2024-07-09, E2024-09-16, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-07-01-A2, E2026-02-03-A3, E2026-06-25-B2

PAT-FLOW-EARLY-BREAK-001 - Balanced early exit from inner search. Exams: E2023-02-07, E2024-02-12, E2025-01-29-A2, E2026-02-03-A2, E2026-06-25-B2

PAT-MEM-BYTE-ARRAY-001 - Signed and unsigned byte arrays. Exams: E2023-02-07, E2023-02-24, E2024-02-12, E2024-02-28, E2024-07-09, E2024-09-16, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2026-02-03-A1, E2026-02-03-A2

PAT-MEM-WORD-ARRAY-001 - Word-array traversal. Exams: E2023-07-04, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-MEM-MATRIX-ROW-MAJOR-001 - Row-major matrix addressing. Exams: E2023-02-24, E2023-05-17, E2023-09-18, E2024-02-12, E2024-02-28, E2024-07-09, E2024-09-16, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-02-12-A1, E2026-02-18-A2, E2026-06-25-B1

PAT-AAPCS-FOUR-ARGS-001 - Four-register argument contract. Exams: E2024-07-09, E2024-09-16, E2026-06-25-B1, E2026-06-25-B2

PAT-AAPCS-STACKED-ARGS-001 - Fifth and later arguments. Exams: E2024-09-16, E2025-07-01-A1, E2025-07-01-A2

PAT-AAPCS-NONLEAF-001 - Non-leaf prologue and epilogue. Exams: E2023-02-07, E2023-02-24, E2023-05-17, E2023-07-04, E2023-09-18, E2024-02-12, E2024-02-28, E2024-07-09, E2024-09-16, E2025-01-29-A1, E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-CPU-FLAGS-001 - APSR flag contract. Exams: E2023-05-17, E2023-09-18, E2024-02-12, E2024-09-16, E2025-01-29-A2, E2025-02-12-A1, E2025-07-01-A1, E2026-02-03-A1, E2026-02-03-A3, E2026-02-18-A2

PAT-CPU-SVC-001 - SVC immediate decoding. Exams: E2023-02-07, E2023-02-24

PAT-CPU-EXCEPTION-FRAME-001 - MSP/PSP exception-frame selection. Exams: E2023-02-24

PAT-TIMER-OWNERSHIP-001 - Timer setup and ownership. Exams: E2023-02-07, E2023-07-04, E2024-02-12, E2024-02-28, E2024-07-09, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-GPIO-EVENT-001 - GPIO event capture. Exams: E2023-02-07, E2023-07-04, E2023-09-18, E2024-02-12, E2024-02-28, E2024-09-16, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1,

E2026-06-25-B1, E2026-06-25-B2

PAT-GPIO-JOYSTICK-001 - Joystick event state machine. Exams: E2025-07-01-A1, E2025-07-01-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-ADC-SAMPLE-001 - ADC sample conversion. Exams: E2023-05-17, E2025-02-12-A1, E2025-02-12-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1

PAT-DAC-STREAM-001 - DAC table streaming. Exams: E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2

PAT-STATE-EVENT-LOOP-001 - ISR-to-foreground event loop. Exams: E2023-07-04, E2023-09-18, E2024-02-12, E2024-02-28, E2024-09-16, E2025-01-29-A3, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-STATE-DEBOUNCE-001 - Deterministic debounce. Exams: E2023-02-07, E2023-09-18, E2024-02-12, E2024-09-16, E2025-01-29-A3, E2025-02-12-A1, E2026-02-03-A1, E2026-02-03-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-TIMER-PERIODIC-001 - Periodic match scheduling. Exams: E2023-07-04, E2024-02-28, E2024-07-09, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-TIMER-FREE-RUNNING-001 - Free-running timer seed. Exams: E2023-02-07, E2024-02-12, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-AAPCS-STACK-SAFETY-001 - Register preservation and stack safety. Exams: E2023-02-07, E2023-02-24, E2023-05-17, E2023-07-04, E2023-09-18, E2024-02-12, E2024-02-28, E2024-07-09, E2024-09-16, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-STATE-IRQ-HANDOFF-001 - Atomic ISR event handoff. Exams: E2023-02-07, E2023-07-04, E2023-09-18, E2024-02-12, E2024-02-28, E2024-09-16, E2025-01-29-A1, E2025-01-29-A2, E2025-01-29-A3, E2025-02-12-A1, E2025-02-12-A2, E2025-07-01-A1, E2025-07-01-A2, E2026-02-03-A1, E2026-02-03-A2, E2026-02-03-A3, E2026-02-18-A1, E2026-02-18-A2, E2026-06-25-B1, E2026-06-25-B2

PAT-TIMER-VECTOR-OWNERSHIP-001 - Exclusive interrupt-vector ownership. Exams: E2023-02-07, E2026-02-18-A1

Adaptation strategy

Keep prototypes and R0-R3 contracts stable. Change constants, configuration, or one named policy block. Rebuild both targets and rerun all related tests after every correction.

Requirements source

The original 1,073-line text specification is authoritative. SHA-256:

5e7c7d51b78173a52eef3e59cd8faa1d0353f8953b9fade6c598d4a904c30ddb. Detailed compliance is maintained in REQUIREMENTS_COMPLIANCE.md.